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 Studierichting: Informatica
 Oefeningenreeks 4A nr. 63.2

$$\begin{aligned}
 I &= \int \sqrt{(1+x^2)^3} dx \\
 &\stackrel{x=\sinh t}{=} \int \sqrt{(1+\sinh^2 t)^3} \cosh t dt = \int \sqrt{\cosh^6 t} \cosh t dt \\
 &= \int \cosh^4 t dt = \int \left(\frac{1+\cosh 2t}{2} \right)^2 dt = \frac{1}{4} \int (1+2\cosh 2t+\cosh^2 2t) dt \\
 &= \frac{t}{4} + \frac{1}{4} \sinh 2t + \frac{1}{8} \int \cosh^2 2t d(2t) \\
 &= \frac{t}{4} + \frac{1}{4} \sinh 2t + \frac{1}{8} \int \frac{1+\cosh 4t}{2} d(2t) \\
 &= \frac{t}{4} + \frac{1}{4} \sinh 2t + \frac{t}{8} + \frac{1}{32} \int \cosh 4t d(4t) \\
 &= \frac{t}{4} + \frac{1}{4} \sinh 2t + \frac{t}{8} + \frac{1}{32} \sinh 4t + C \\
 &\stackrel{t=\operatorname{Bgsinh} x}{=} \frac{3 \operatorname{Bgsinh} x}{8} + \frac{1}{4} \sinh (2 \operatorname{Bgsinh} x) + \frac{1}{32} \sinh (4 \operatorname{Bgsinh} x) + C
 \end{aligned}$$

References